Evaluating Generative Models

Advanced Natural Language Processing
Sukkur IBA University
Sher Muhammad Daudpota

Models’ performance evaluation

- Why accuracy, precision, recall, AUC etc. are not good metrics to assess generative models performance?

BLUE, ROUGE, NGrams

- BLUE (Bilingual Evaluation Understudy)
  - For language translation
- ROUGE
  - Mainly for evaluating generated summaries

N-Grams

“The Cat Sleeps Quietly”

One-Grams?

Two-Grams?

N-Grams

“The Cat Sleeps Quietly”

One-Grams: “The”, “Cat”, “Sleeps”, “Quietly”

Two-Grams: “The”, “Cat”, “The Cat”, “Sleeps”, “Cat Sleeps”, “Quietly”, “Sleeps Quietly”

Why discussing N-Grams

What N-gram-based metrics usually do is compare the n-grams of the ***generated sentence***, with those of a ***reference text*** and indicate whether,

1. there is a lot
2. a little,
3. No difference between the generated text and the reference text.

Measuring the translation quality with BLEU

- Measures the similarity of the generated translation to the reference translations provided

Measuring the translation quality with BLEU

```
#Sentences to Translate.
sentences = [
    "In the previous chapters, you've mainly seen how to work with OpenAI
    models, and you've had a very practical introduction to Hugging Face's
    open-source models, the use of embeddings, vector databases, and
    agents.",
    "These have been very practical chapters in which I've tried to
    gradually introduce concepts that have allowed you, or at least I hope
    so, to scale up your knowledge and start creating projects using the
    current technology stack of large language models."]

#Spanish Translation References.
reference_translations = [
    ["En los capítulos anteriores has visto mayoritariamente como trabajar
    con los modelos de OpenAI, y has tenido una introducción muy práctica
    a los modelos Open Source de Hugging Face, al uso de embeddings, las
    bases de datos vectoriales, los agentes."],
    ["Han sido capítulos muy prácticos en los que he intentado ir
    introduciendo conceptos que te han permitido, o eso espero, ir
    escalando en tus conocimientos y empezar a crear proyectos usando el
    stack tecnológico actual de los grandes modelos de lenguaje."]
    ]
```

Measuring the translation quality with BLEU

```
reference_translations = [
    ["reference translation 1 for text 1", "reference translation 2 for
    text 1", ...],
    ["reference translation 1 for text 2", "reference translation 2 for
    text 2", ...],
    ...
]
```

BLEU: *bilingual evaluation understudy*

- Useful for text generation quality assessment
- Especially machine translation
- BLEU-1, BLEU-2, BLEU-3.. Where 1, 2, 3.. are n-grams

BLEU: *bilingual evaluation understudy*

BLEU = sum of clipped count / sum of unigram in candidate text

Clipped count = min (Count, max frequency in reference)

- **Count:** The number of times an n-gram appears in the candidate translation.
- **Clipped Count:** The number of times an n-gram appears in the reference translation, clipped to the maximum number of times it appears in any reference.

BLEU: *bilingual evaluation understudy*

- Example:

Reference 1: the cat is on the mat

Reference 2: there is a cat on the mat

Candidate: the cat the cat on the mat

BLEU: *bilingual evaluation understudy*

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Count: The number of times an n-gram appears in the candidate translation.
Clipped Count: The number of times an n-gram appears in the reference translation, clipped to the maximum number of times it appears in any reference.

Unique Unigram	Count	Clipped Count
the	3	2
cat	2	1
on	1	1
mat	1	1

BLEU: *bilingual evaluation understudy*

Example (BLUE-1):

Reference 1: the cat is on the mat

Reference 2: there is a cat on the mat

Candidate: the cat the cat on the mat

Unique Unigram	Count	Clipped Count
the	3	2
cat	2	1
on	1	1
mat	1	1

BLEU = sum of clipped count / sum of unigram in candida

BLEU = 5 / 7 = 0.714

BLEU: *bilingual evaluation understudy*

Example (BLUE-2):

Reference 1: the cat is on the mat

Reference 2: there is a cat on the mat

Candidate: the cat the cat on the mat

BLEU = sum of clipped count / sum of unigram in candidate text

BLEU = 4 / 6 = 0.67

Unique Bigram	Count	Clipped Count

BLEU: *bilingual evaluation understudy*

Example (BLUE-2):

Reference 1: the cat is on the mat

Reference 2: there is a cat on the mat

Candidate: the cat the cat on the mat

BLEU = sum of clipped count / sum of unigram in candidate text

BLEU = 4 / 6 = 0.67

Unique Unigram	Count	Clipped Count
the cat	2	1
cat the	1	0
cat on	1	1
on the	1	1
the mat	1	1

ROUGE: *Recall-Oriented Understudy for Gisting Evaluation*

- Rouge-1, 2, 3.. Correspond to n-grams
- Rouge-L uses longest common sequence to calculate the score
- Rouge uses ***recall***
- BLEU uses precision

ROUGE: *Recall-Oriented Understudy for Gisting Evaluation*

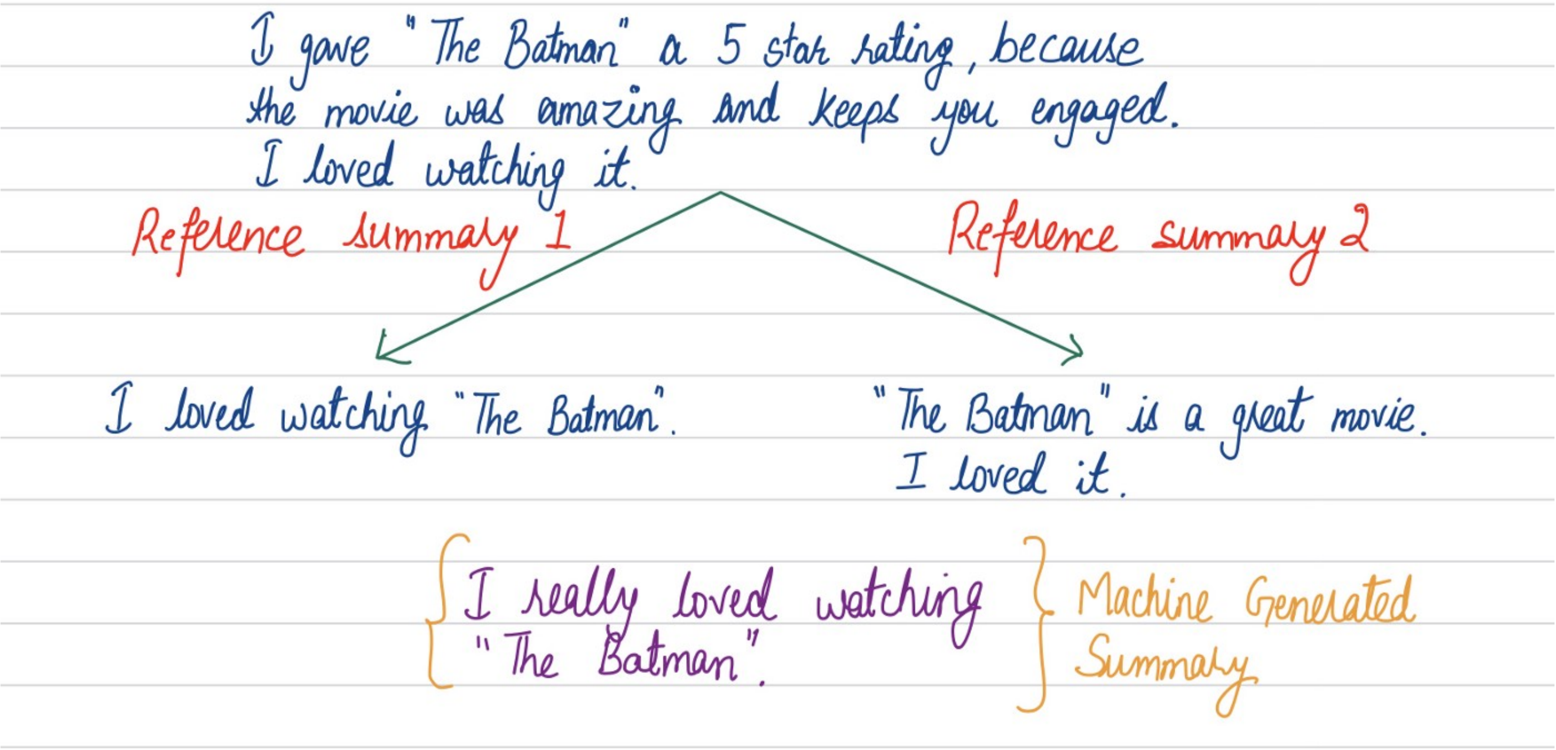
- BLEU vs Roughe Difference:
  - **Rouge calculates how much the n-gram in the *human reference text* appeared in the *machine generated text***
  - **BLEU calculates how much the n-gram in the *machine generated text* appeared in the *human reference text***

ROUGE: *Recall-Oriented Understudy for Gisting Evaluation*

- Rouge is usually used in text summarization

Rouge Example

Based on https://levelup.gitconnected.com/performance-analysis-of-text-summary-generation-models-using-rouge-score-72e4a8418df6



Rouge Example

Machine Generated:

I really loved watching "The Batman".

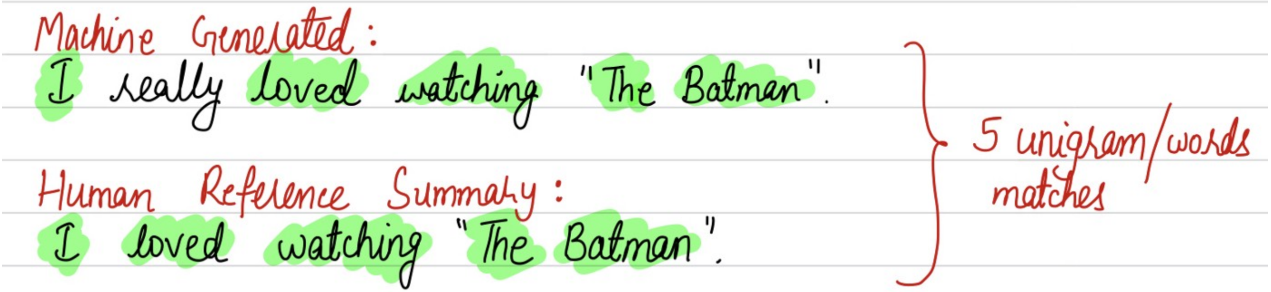
Human Reference Summary:

I loved watching "The Batman".

5 unigram/words matches

Rouge Example

- Rouge Recall = Word in both summaries / total words in reference



- Rough Recall = 5 / 5 = 1

Rouge Example

- How about a bad summary like,

“*I really really really really really loved watching “The Batman.”*

Rouge Example

- How about a bad summary like,

Reference: I love watching “The Batman”

Machine Generated: *I really really really really really loved watching “The Batman.”*

Recall = 5 / 5 = 1

Rouge Example

- Which indicates recall is not sufficient….
- Here comes Rouge Precision

Rouge Example

- Precision = Word in both summaries / total words in generated summary

Reference: I loved watching The Batman

Machine Generated: *I really really really really really loved watching The Batman.*

Recall = 5 / 5 = 1

Precision = 5 / 10 = 0.5

f1-score = (2 x precision x recall) / (precision + recall)

f1-score = 1 / 1.5 = 0.66

Rouge-2

Human: I loved watching the batman

Generated: I really loved watching the batman

Human Bigram	Generated Bigram

Rouge-2

Human: I loved watching the batman

Generated: I really loved watching the batman

Human Bigram	Generated Bigram
I loved	I really
loved watching	really loved
watching the	loved watching
the batman	watching the
	the batman

Rouge-2

Human: I loved watching the batman

Generated: I really loved watching the batman

Recall = 3 / 4 = 0.75

Precision = 3 / 5 = 0.6

F1 = 2 * 0.75 * 0.6 / 0.75 + 0.6 = 0.66

Human Bigram	Generated Bigram
I loved	I really
loved watching	really loved
watching the	loved watching
the batman	watching the
	the batman

Rouge-L

- treats each summary as a sequence of words
- looks for the longest common subsequence (LCS)

Machine Generated:

I really loved watching "The Batman".

Human Reference Summary:

I loved watching "The Batman".

longest common subsequence will be:

"I loved watching The Batman".

METEOR: *Metric for Evaluation of Translation with Explicit Ordering*

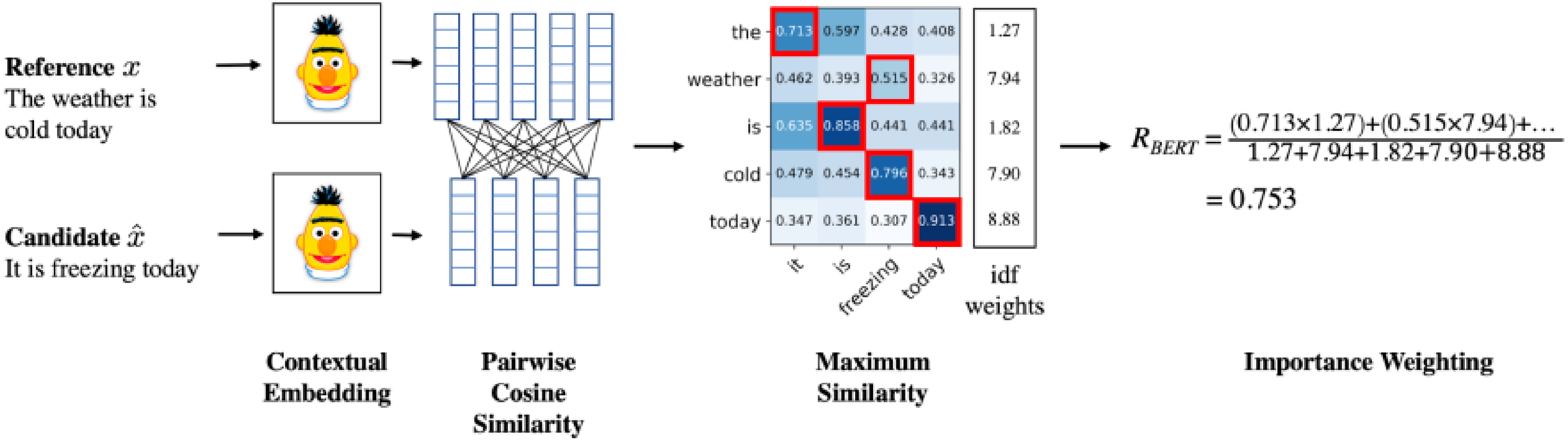
- Incorporates the harmonic mean of the precision and recall
- Also considers synonyms for word matching
- The equation uses the *precision* and *recall,* combined using the harmonic mean with recall weighted 9 times more than precision.

Model based metrics

- N-gram based disadvantage,
    - Word matching only
    - No semantic consideration
- Model based solve the issue,
    - BERTScore
    - BLEURT

Model based metrics

- BERTScore



Code: LLM_Performance_Evaluation.ipynb

Evaluating LLM Summaries Using Embedding Distance with LangSmith

- Uses cosine distance between embeddings to identify which summary is most similar to the original

- Embedding is nothing more than a vector that tries to capture the semantic meaning of the converted text

- To calculate the distance between the embeddings, Langsmith uses, by default, cosine distance. This metric can take a value between 0 and 2

Evaluating Language Models with Language Models

```
Score the following translation from {source_lang} to
{target_lang} with respect to the human reference on a continuous scale
from 0 to 100 that starts with "No meaning preserved", goes through "Some
meaning preserved", then "Most meaning preserved and few grammar mistakes",
up to "Perfect meaning and grammar".

{source_lang} source: "{source_seg}"
{target_lang} human reference: "{reference_seg}"
{target_lang} translation: "{target_seg}"

Score (0-100):
```